

GREYBULL, WY, USA

WMO: 726667

Lat: 44.517N

Lon: 108.082W

Elev: 1199

StdP: 87.73

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 24048

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.2	(c) -21.8	(d) -27.8	(e) 0.3	(f) -24.9	(g) -24.3	(h) 0.5	(i) -21.0	(j) 13.9	(k) -3.8	(l) 11.7	(m) -3.6	(n) 1.0	(o) 130	(p) 0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.5	(c) 36.3	(d) 16.8	(e) 34.7	(f) 16.3	(g) 32.9	(h) 15.9	(i) 18.7	(j) 31.2	(k) 17.6	(l) 30.0	(m) 16.8	(n) 29.5	(o) 3.6	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 14.3	(b) 11.8	(c) 22.0	(d) 13.1	(e) 10.9	(f) 21.2	(g) 12.2	(h) 10.2	(i) 20.7	(j) 57.7	(k) 30.3	(l) 53.9	(m) 29.8	(n) 51.2	(o) 29.4	(p) 25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	10.9	9.0	-30.6	39.0	3.5	1.6	-33.2	40.2	-35.2	41.1	-37.2	41.9	-39.7	43.1
			-30.8	19.7	3.5	2.0	-33.3	21.1	-35.3	22.3	-37.2	23.4	-39.7	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-8.1	-5.3	3.4	8.2	13.6	19.5	24.4	22.3	16.3	8.2	0.0	-7.4		
	DBStd	12.15	6.08	6.80	5.61	4.53	4.61	4.10	2.75	3.08	4.29	4.79	6.40	5.95		
	HDD10.0	2235	560	428	211	84	18	1	0	0	5	88	301	539		
	HDD18.3	4197	818	661	464	303	158	34	1	7	87	315	551	797		
	CDD10.0	1507	0	0	5	32	129	285	447	383	193	33	1	0		
	CDD18.3	426	0	0	0	1	11	68	189	131	25	1	0	0		
	CDH23.3	6368	0	0	2	29	248	1017	2638	1841	558	33	0	0		
	CDH26.7	3134	0	0	0	4	72	457	1448	928	218	6	0	0		
(14) Wind	WSAvg	3.0	1.9	2.5	3.4	4.1	3.9	3.6	3.2	3.0	2.9	2.9	2.5	2.2		
(15) Precipitation	PrecAvg	177	6	6	10	19	31	33	17	9	18	20	7	6		
	PrecMax	251	20	18	25	57	65	82	68	32	47	45	20	17		
	PrecMin	124	0	0	1	5	7	10	0	2	2	5	0	0		
	PrecStd	38	5	4	6	11	18	21	17	7	13	12	5	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	13.2	22.3	26.6	31.5	37.0	38.7	37.4	34.0	27.0	18.1	9.0		
		MCWB	3.1	5.2	9.4	11.3	14.4	16.2	17.4	18.0	15.5	12.7	8.3	3.6		
	2%	DB	4.9	9.6	18.7	23.3	28.7	34.1	37.1	35.4	31.8	23.2	14.7	5.8		
		MCWB	0.9	3.7	7.8	10.0	13.4	16.1	17.9	16.0	14.8	11.2	6.5	1.6		
	5%	DB	2.6	6.8	16.0	20.6	26.3	31.9	35.5	33.7	29.1	20.3	12.0	3.0		
		MCWB	-0.4	1.8	6.6	9.3	12.9	15.4	16.7	15.9	13.6	10.2	5.1	-0.3		
	10%	DB	0.6	3.9	12.8	17.7	23.3	29.1	33.7	32.1	26.9	17.3	8.8	1.1		
		MCWB	-1.7	0.3	5.3	8.0	11.6	14.8	16.4	15.6	13.1	8.8	3.7	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	6.2	10.1	12.3	15.7	18.5	21.2	20.1	17.1	13.6	8.8	4.2		
		MCDB	7.5	12.0	20.8	24.7	27.6	28.6	36.6	34.6	27.8	24.2	16.9	8.2		
	2%	WB	1.3	4.0	8.3	10.6	14.5	17.1	19.2	18.0	15.7	12.0	6.8	1.7		
		MCDB	4.5	9.1	17.6	21.0	25.5	29.7	32.7	30.5	27.5	21.1	13.9	5.1		
	5%	WB	-0.1	2.1	6.9	9.6	13.5	16.2	18.1	17.0	14.7	10.7	5.4	0.1		
		MCDB	2.4	6.1	15.2	19.3	23.8	29.2	30.7	29.3	26.6	19.0	11.0	3.0		
	10%	WB	-1.7	0.5	5.6	8.5	12.5	15.3	17.3	16.2	13.6	9.4	3.9	-1.3		
		MCDB	0.5	3.8	12.4	17.4	21.7	27.2	29.8	28.6	24.8	16.6	8.5	1.0		
(35) Mean Daily Temperature Range	MDBR		12.4	12.4	14.2	14.1	13.9	15.4	17.5	17.4	16.7	14.1	13.3	11.8		
	5% DB	MCDBR	13.8	15.0	19.3	19.5	18.5	19.6	20.1	20.4	21.6	19.9	17.4	13.5		
		MCWBR	10.3	9.6	10.3	9.1	7.1	6.5	6.7	7.3	8.5	9.9	10.4	9.9		
	5% WB	MCDBR	13.4	14.4	17.7	17.8	16.9	17.9	18.0	17.7	18.8	18.2	16.3	13.1		
		MCWBR	10.2	9.6	9.7	8.4	6.5	6.2	6.8	6.8	7.6	9.4	9.9	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.249	0.254	0.279	0.317	0.339	0.344	0.363	0.358	0.325	0.286	0.260	0.245		
	taud		2.523	2.525	2.480	2.369	2.374	2.386	2.365	2.359	2.442	2.530	2.532	2.527		
	Ebn at Noon		903	958	967	946	930	922	900	893	900	903	879	873		
	Edh at Noon		65	78	94	114	118	117	118	115	97	77	64	58		
(44) All-Sky Solar Radiation	RadAvg		1.78	2.76	4.06	5.16	6.04	7.13	7.23	6.15	4.84	3.13	2.01	1.45		
	RadStd		0.13	0.18	0.28	0.26	0.55	0.47	0.28	0.27	0.33	0.24	0.17	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page